# Predicting the Regulatory Impact of Non-Coding Single Nucleotide Variants with Machine Learning

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## Abstract

The use of machine learning in biology continues to be transformative as we apply this technology to personalized medicine. The project aims to utilize machine learning to predict the regulatory impact of noncoding single nucleotide variants (SNV) through functional genomic data. To build a dataset for training and testing, melanoma related SNV information was taken from NCBI variant viewer and functional genomic data from Enricher. This final multimodal dataset of histone modifications, transcription factor binding data, local DNA sequence context, and clinical significance of melanoma for each intronic SNV was used for training on a logistic regression model. After initial training for binary classification of variants between benign or pathogenic, the model achieved an accuracy of 62.5% on the test set but only 33.33% on an independent dataset. These results indicate a need for further training with a larger and more diverse training dataset. A more generalized model applicable across different datasets can capture complex feature patterns among non-coding single nucleotide variants facilitating the advancement of more precise, individualized medicine.


**Keywords:** machine learning, logistic regression, binary classification, personalized medicine, melanoma, single nucleotide variant, non-coding variant

## Introduction

Non-coding intronic variants do not directly impact the final protein structure of a gene but may still have regulatory effects that cause disease. These regulatory effects include, but not limited to, alternative splicing to create distinct protein isoforms and mechanisms that increase or decrease expression of genes [4]. The impact of introns in gene expression is still being studied as our knowledge on them is still incomplete [1]. Understanding the impact of intronic variants is important to our advancement of medicine as we further realize their effects on our genome. In melanoma, certain non-coding single nucleotide variants (SNVs) are known to be cancer causing. However, not all SNVs are documented with certain significance on the progression of a disease and not every single SNVs can be investigated for its molecular interactions as there are many SNVs for a gene. Melanoma is a disease with many variants documented with known and unknown impact. This project focuses on melanoma to train a logistic regression machine learning algorithm to predict whether specific SNVs, given regulatory impact, are benign or disease causing. Utilizing a machine learning algorithm to predict the effects of SNVs can improve our ability to characterize the regulatory consequences of certain SNVs, since many variants lack clearly documented clinical significance [10]. A generalized machine learning model can be applied across similar diseases to allow for more targeted and efficient research, as it can identify meaningful patterns within complex functional genomic data.

## Data Sets

To compile the training dataset and independent testing dataset, variants were selected with NCBI Variation Viewer focusing on melanoma associated non-coding variants from dbSNP (public database of single nucleotide polymorphisms), ClinVar (public database of genetic variations and phenotypes), and GWAS (genome wide association studies) Catalog. In this process, three genes were chosen for training: TYR, SLC45A2, and CDKN2A. For each gene, 10 non-coding SNVs were identified. For the independent testing dataset, the MC1R gene was chosen and 6 SNVs were identified.

In NCBI Variation Viewer, the following criteria was applied for variant selection: in dbSNP, in ClinVar, a single nucleotide variant, an intron variant, from genomic regions associated in GWAS, and has reference and alternate alleles (ref/alt). The following was recorded for each variant into the dataset from NCBI Variation Viewer: chromosome number, gene name, rsID, position, and ref/alt alleles. The label for variant-condition record (RCV) from ClinVar is displayed for each variant in NCBI Variation Viewer and recorded into the dataset. This label for clinical significance is labeled 1 for Pathogenic, Likely-Pathogenic, or Uncertain Significance (VUS) and 0 for Benign or Likely-Benign in the dataset. The distance of the variant to the transcription start site was recorded. Depending on this distance, the variant is labeled as a promoter, enhancer, or none. Closer distances are labeled as 1 in is_promoter. Longer distances are labeled as 1 in is_enhancer. Even longer distances are labeled as none with values in is_promoter and is_enhancer as 0. From dbSNP in NCBI, the upstream and downstream sequences of 25 base pairs adjacent to the variation were recorded for each SNV. These sequences were used to calculate the GC content as a feature class for training the model because GC content can impact the binding of transcription factors.

Histone modification data, as an aspect of functional genomics, were assembled using Enricher from the ENCODE Histone Modifications 2015 dataset. The histone modifications from Enricher were recorded: H3K27me3 related to inactive regions of the genome and Polycomb repression [2], H3K9me3 related to heterochromatin preventing transcription factors binding for transcription [9], H3K36me3 related to transcription of gene bodies [9], and H3K79me2 related to actively transcribed chromatin [3].

Transcription factor (TF) binding data was extracted from ENCODE TF ChIP-seq 2015 using Enricher. The specific TF binding data were recorded for each gene: E2F4 related to increased activity for melanoma cells [8]; MYC related to driving aggressive melanoma [7]; EZH2, SUZ12, CBX8 related to Polycomb complex formation [2]; and GABPA, TCF12, FOSL1 linked to enhancer activation and activation of promoter in melanoma.

The same histone modification and transcription factor binding scores were assigned for all SNVs that were from the same gene, approximating the conditions for those variants. This was due to the fact that ChIP-seq data specific to melanoma variants were difficult to attain. Histone modification data and TF binding data are important feature classes for training a machine learning model because SNVs at TF binding sites or with certain histone modifications can impact gene expression and transcription. Patterns extracted from these feature datasets can improve prediction performance from the machine learning model.

## Approach

This project is split into 4 parts: dataset creation, data cleaning and transformation, machine learning training, and machine learning testing. Information for single nucleotide variants were compiled in the manner as described in the *Data Sets* section above utilizing NCBI Variation Viewer, ClinVar, and Enricher into a comma-separated values (csv) document. Python was chosen as the language for the code. This csv file was extracted into a Pandas dataframe and further transformation was performed to prepare a feature matrix and target vector ready for training the machine learning model. A logistic regression model was used to generate functional scores for each variant to estimate the likelihood of impact as benign or pathogenic, following the labeling convention from ClinVar. Before the model was trained, the feature matrix was further transformed by converting reference and alternate nucleotides for each SNV with one-hot encoding as they needed to be converted to continuous variables for logistic regression [5]. The feature matrix and target vector were split into a training and testing set. The training set first fit the logistic regression model from scikit-learn, and the testing set was then used to predict the labels as 0 for benign or 1 for pathogenic/uncertain. Probabilities for these predictions were also generated as likelihood of impact by SNVs on melanoma. The trained model was further tested with the independent testing dataset of gene MC1R. Performance analysis of the logistic regression model was completed with an accuracy score, confusion matrix, classification report, and area under the Receiver Operating Characteristic curve (AUC–ROC).

## Methods

Dataset creation follows the same detailed illustration in the *Data Sets* section by utilizing NCBI Variation Viewer, Enricher, and ClinVar. This project utilized NCBI Variation

Viewer to search for specific SNVs associated with melanoma. As mentioned in the *Data Sets* section, 10 intronic SNVs were chosen for each of the 3 genes and 6 intronic SNVs were chosen for the MC1R gene. From NCBI Variation Viewer, chromosome number, gene name, rsID, position, ref/alt alleles, and the label for clinical significance from ClinVar were recorded into the final csv file. The top scores for histone modification and transcription factor binding were recorded from Enricher by individually searching for the genes chosen. The datasets used in Enricher are ENCODE Histone Modifications 2015 and ENCODE TF ChIP-seq 2015, respectively. Flanking sequences of 25 base pairs upstream and downstream of the SNV were recorded from dbSNP on NCBI by searching for the specific SNV recorded from NCBI Variation Viewer. This final dataset is stored in the data.csv file.

Computation was completed in the Python programming language and stored in the final.py file. A conda virtual environment was created with all necessary dependencies in the environment.yml file. In the initial series of data cleaning and transformation, the Pandas library was used to extract the csv file of the final dataset into a data frame. This data frame was transformed by extracting each individual histone modification and transcription binding data point with a regular expression function, as all data points in the raw data were initially entered into a single cell. GC content was calculated programmatically with the recorded flanking sequences while making sure to remove empty spaces in the raw data. The training dataset of three genes (TYR, SLC45A2, CDKN2A) each with ten SNVs was separated from the independent testing dataset of the MC1R gene and the six SNVs. To prepare the dataset further for model fitting, the OneHotEncoder function from scikit-learn was applied to ref/alt nucleotides and extracted into separate numerical features each representing different reference or alternate nucleotides.

To prepare the logistic regression model for machine learning training, functions from scikit-learn were utilized. The Pipeline function was used for data processing to standardize the feature data with StandardScaler function and a logistic regression model was instantiated with the LogisticRegression function. With the train_test_split function, the training dataset with genes TYR, SLC45A2, and CDKN2A was automatically split into a test set size of 25% and training set size of 75%. This training set was fitted into the logistic regression model with the fit function. Afterwards, the predict function and predict_proba function was used to predict the labels, benign or pathogenic and generate likelihood of impact by the SNVs on melanoma with the test set. To compare the predictions and observations, the test set was merged with the original training dataset to identify the SNV associated with the predications, since nonnumerical identifiers were removed when fitting the model. The same processes here with the training dataset are completed for the independent testing dataset of the MC1R gene as well.

Performance analysis for the evaluation of the logistic regression model was completed by generating scores and graphs with functions from scikit-learn separately for both the training dataset and the independent testing dataset. Using the accuracy_score and confusion_matrix functions, the accuracy was calculated for predictions by the model and a confusion matrix was generated. The confusion matrix was further evaluated by transforming it into a heatmap with the seaborn library and matplotlib library. A classification report of precision, recall, and F-1 score was generated for the predictions. From scikit-learn, the roc_auc_score and roc_curve functions were used to further evaluate the model and calculate the AUC score for the predictions. From the matplotlib library, the true positive rates, false positive rates, and thresholds were taken from the roc_curve function to plot an AUC–ROC curve. All plots are stored in the output_plots directory, and the terminal log containing the scores is saved in final_code_results.txt.

## Results and Discussion

From the terminal log, the results for the predictions, likelihood measurements for the predictions, accuracy of the model, confusion matrix, classification report, and ROC-AUC score are presented below and could also be found in final_code_results.txt.

In Fig. 1 for the initial model using genes TYR, SLC45A2, and CDKN2, we can see that each prediction in "pred_label" of either 0 for likely benign and 1 for likely pathogenic/uncertain generates a likelihood measurement in "Likelihood_Pathogenic/Uncertain Sig". Unexpectedly, we can see that the model predicted two labels incorrectly with the top two highest confidences in row number 2 and 5. As we can see in the confusion matrix in Fig. 2, three labels were predicted incorrectly out of eight predictions. All errors in this evaluation were in predicting 1 when the true label was 0. No errors were made in predicting 0 when the true label was 1. The accuracy score of this model returned to be 62.50% because we have 5 true positives and negatives out of 8 total predictions. This could be attributed to an unequal distribution of genes in the training and testing set as only one label came from SLC25A2 in the test set (Fig. 1).

```
---------- Initial Model Training and Evaluation ----------

Comparing true labels with predicted labels,
    including the predicted likelihood of pathogenicity/uncertain significance (Initial Training):
       gene    Intron_SNV chr         pos ref alt  true_label  pred_label  Likelihood_Pathogenic/Uncertain Sig
0       TYR    rs61754381  11  89227816.0   T   A         1.0         1.0                              0.657802
1    CDKN2A  rs1563886124   9  21968298.0   C   A         0.0         1.0                              0.540052
2    CDKN2A   rs777951032   9  21968259.0   G   A         0.0         1.0                              0.939565
3    CDKN2A   rs866937105   9  21970897.0   C   A         1.0         1.0                              0.687451
4       TYR  rs2496552201  11  89191181.0   A   G         1.0         1.0                              0.748501
5    CDKN2A   rs772321849   9  21968257.0   A   T         0.0         1.0                              0.882723
6   SLC45A2   rs779605677   5  33951686.0   G   C         0.0         0.0                              0.240258
7       TYR   rs765098311  11  89284759.0   C   A         0.0         0.0                              0.062506
```

Fig. 1. Likelihood of predictions and predictions of initial model using genes TYR, SLC45A2, and CDKN2.

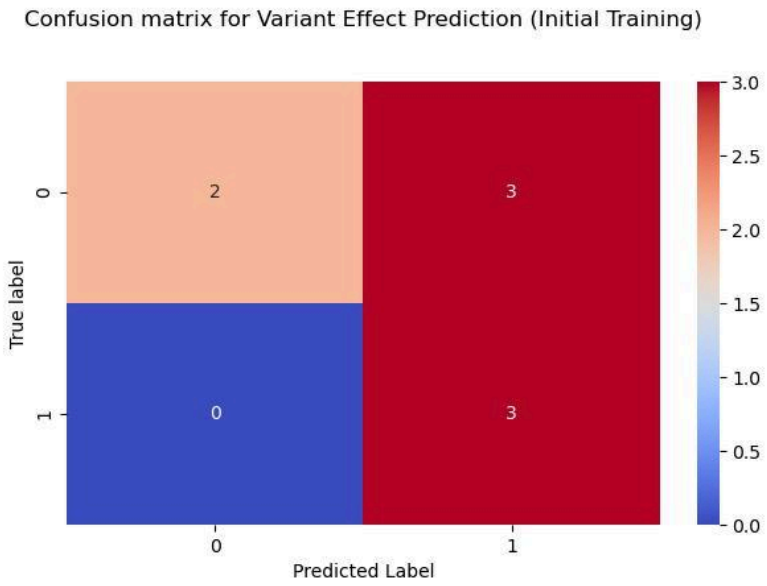


Fig. 2. Confusion matrix of initial model using genes TYR, SLC45A2, and CDKN2.

In the classification report of this initial model (Fig. 3), precision was 0.5 indicating a weak ability of the model to predict likely pathogenic or uncertain variants. However, recall was 1.0, indicating a strong ability to correctly identify all true likely pathogenic or uncertain variants without missing any true positives. A moderate F1 score of 0.67 in this model also indicates the limited ability to minimize false positives and false negatives (Fig. 3). With a small training set, a low accuracy and imbalance between precision and recall is expected.

```
Classification Report (Initial Training):
                                                          precision    recall  f1-score   support

                                 Benign & Likely-Benign: 0       1.00      0.40      0.57         5
Pathogenic, Likely-Pathogenic, Uncertain-Significance (VUS): 1       0.50      1.00      0.67         3

                                                 accuracy                           0.62         8
                                                macro avg       0.75      0.70      0.62         8
                                             weighted avg       0.81      0.62      0.61         8
```

Fig. 3. Classification Report of initial model using genes TYR, SLC45A2, and CDKN2.

The Receiver Operating Characteristic (ROC) curve returned an area under the curve (AUC) score of 0.6, indicating weak classification performance (Fig. 4).

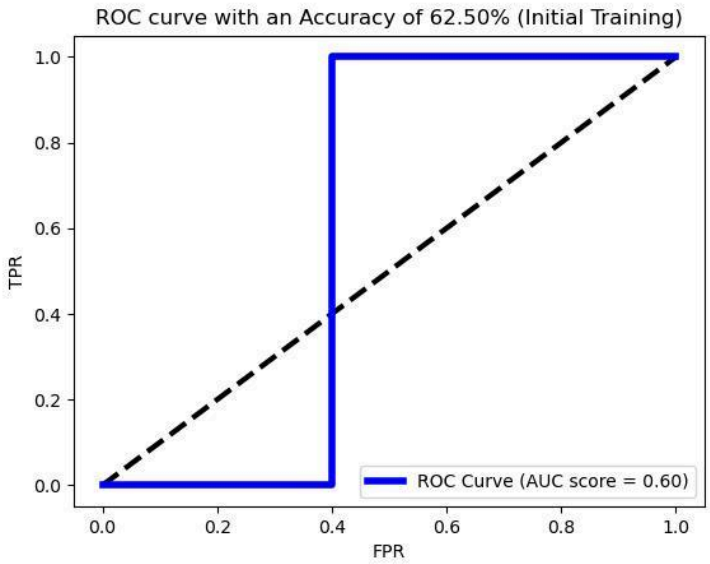


Fig. 4. ROC curve of initial model using genes TYR, SLC45A2, and CDKN2.

During independent testing of the model using the MC1R gene, the AUC score decreases even more to 0.5 (Fig. 5), indicating random classification. This demonstrates that the initial model lacks generalization to external datasets with different histone modifications and transcription binding features.

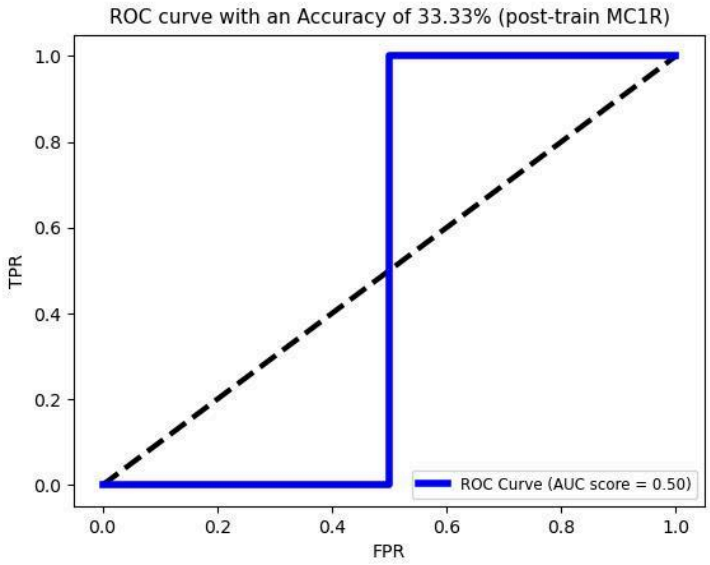


Fig. 5. ROC curve testing model using MC1R gene.

In Fig. 6 involving the independent testing dataset of the MC1R gene, we can again see that each prediction in “pred_label” of either 0 for likely benign and 1 for likely pathogenic/uncertain generates a likelihood measurement in “Likelihood_Pathogenic/Uncertain Sig”. Unexpectedly, the model made more prediction errors predicting 0 when the true label was 1 than vice versa. As we can see in the confusion matrix in Fig. 7, four labels were predicted incorrectly out of six total predictions. Only two predictions were correct leading to an accuracy score of 33.33% because we have one true positive and one true negative out of six total predictions. This could be attributed to a small training set in the initial model and limited diversity of features in the training dataset.

```
---------- Using MC1R on Built Model for Independent Testing ----------

Comparing true labels with predicted labels,
    including the predicted likelihood of pathogenicity/uncertain significance (post-train MC1R):
    gene   Intron_SNV chr         pos ref alt  true_label  pred_label  Likelihood_Pathogenic/Uncertain Sig
0  MC1R  rs868197501  16  89918659.0   C   T         1.0         0.0                              0.217898
1  MC1R  rs553184868  16  89918755.0   G   A         1.0         0.0                              0.365028
2  MC1R    rs3212379  16  89918723.0   C   T         0.0         1.0                              0.761812
3  MC1R  rs886052500  16  89918751.0   C   T         1.0         0.0                              0.190559
4  MC1R  rs932918344  16  89918692.0   C   T         1.0         1.0                              0.596129
5  MC1R    rs3212359  16  89918769.0   C   T         0.0         0.0                              0.136046
```

Fig. 6. Likelihood of predictions and predictions testing model using MC1R gene.

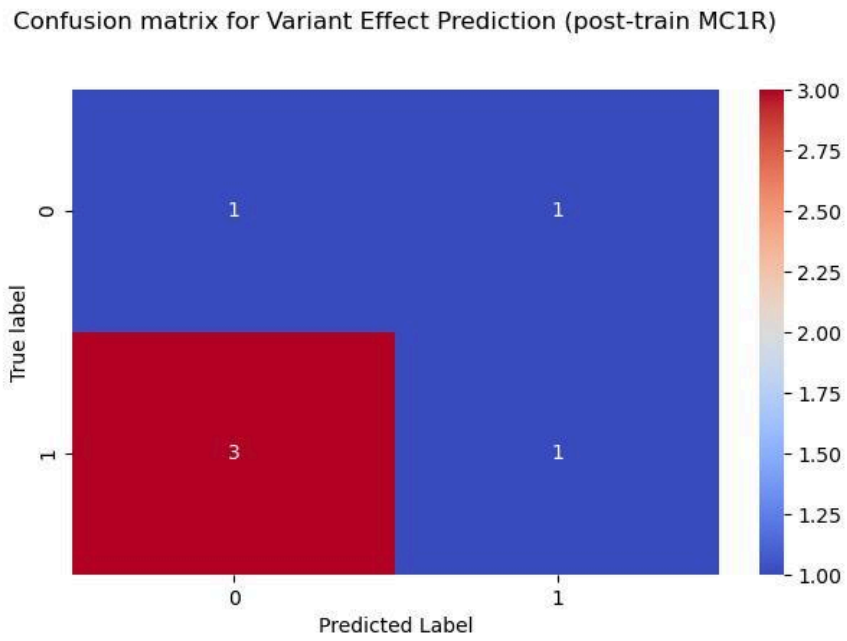


Fig. 7. Confusion matrix testing model using MC1R gene.

In the classification report of this initial model (Fig. 8), precision was 0.5 indicating a weak ability of the model to predict likely pathogenic or uncertain variants. However, recall was even worse at 0.25, indicating a weaker ability to correctly identify all true likely pathogenic or uncertain variants without missing any true positives. A very low F1 score of 0.33 in this model indicates very limited ability to minimize false positives and false negatives (Fig. 8). With a small initial training set for the model and differing histone modification and TF binding features, a low accuracy and low F1 score is likely.

```
Classification Report (post-train MC1R):
                                                          precision    recall  f1-score   support

                                  Benign & Likely-Benign: 0      0.25      0.50      0.33         2
Pathogenic, Likely-Pathogenic, Uncertain-Significance (VUS): 1      0.50      0.25      0.33         4

                                                  accuracy                           0.33         6
                                                 macro avg      0.38      0.38      0.33         6
                                              weighted avg      0.42      0.33      0.33         6
```

Fig. 8. Classification Report testing model using MC1R gene.

## Conclusion

Accuracy and evaluation scores were higher for the initial training of the model. However, subsequent testing with an independent dataset yielded much lower accuracy from 62.5% initially to 33.3% in the independent testing. Recall, and F1 scores also significantly decreased, but precision stayed the same. Nonetheless, because of the small initial testing dataset and smaller independent testing dataset, these performance metrics should be interpreted with caution as they are highly variable. Further training with more diverse histone modification and TF binding features could yield a more generalizable model. Increasing the number of SNVs from a broader range of genes would provide the model with more comprehensive information to potentially improve prediction performance. Future testing should aim to utilize more balanced training and testing datasets ensuring more uniform representation of genes with respect to SNV distribution.

## Team Member Contributions

**Other Teammembers:**

The initial dataset creation was completed in collaboration with Lawrence. However, the final dataset was retrieved from Jane and Zhanmin and was utilized as a .csv file in the fulfilment of this project.

**Lawrence:**

All work explained in significant detail in this report, excluding the dataset creation in the data.csv file, was completed solely by Lawrence. This work includes data cleaning and transformation in Pandas; model training, testing, and performance analysis; and writing the project report.

## References


[1] A. B. Rose, "Introns as Gene Regulators: A Brick on the Accelerator," *Frontiers in Genetics*, vol. 9, no. 672, Feb. 2019, doi: https://doi.org/10.3389/fgene.2018.00672.

[2] A. Piunti and A. Shilatifard, "The roles of Polycomb repressive complexes in mammalian development and cancer," *Nature Reviews Molecular Cell Biology*, vol. 22, Mar. 2021, doi: https://doi.org/10.1038/s41580-021-00341-1.

[3] A. T. Nguyen and Y. Zhang, "The diverse functions of Dot1 and H3K79 methylation," *Genes & Development*, vol. 25, no. 13, pp. 1345–1358, Jul. 2011, doi: https://doi.org/10.1101/gad.2057811.

[4] B.-S. Jo and S. S. Choi, "Introns: The Functional Benefits of Introns in Genomes," *Genomics & Informatics*, vol. 13, no. 4, p. 112, 2015, doi: https://doi.org/10.5808/gi.2015.13.4.112.

[5] G. Zeng, "On the analytical properties of category encodings in logistic regression," *Communications in Statistics - Theory and Methods*, vol. 52, no. 6, pp. 1870–1887, Jun. 2021, doi: https://doi.org/10.1080/03610926.2021.1939382.

[6] J. Tiche, "Predicting the Impact of Non-coding Genetic Variants on Gene Expression," San Jose State University, Dec. 2026.

[7] M. F. Zacarías-Fluck *et al.*, "Reducing MYC's transcriptional footprint unveils a good prognostic gene signature in melanoma," *Genes & Development*, vol. 37, no. 7–8, pp. 303–320, Sep. 2025, doi: https://doi.org/10.1101/gad.352999.125.

[8] R. Halaban, E. Cheng, Y. Smicun, and J. Germino, "Deregulated E2f Transcriptional Activity in Autonomously Growing Melanoma Cells," *Journal of Experimental*

*Medicine*, vol. 191, no. 6, pp. 1005–1016, Mar. 2000, doi: https://doi.org/10.1084/jem.191.6.1005.

[9] R. Nakato and T. Sakata, “Methods for ChIP-seq analysis: A practical workflow and advanced applications,” *Methods*, vol. 187, Mar. 2020, doi: https://doi.org/10.1016/j.ymeth.2020.03.005.

[10] W. Burke, E. Parens, W. K. Chung, S. M. Berger, and P. S. Appelbaum, “The Challenge of Genetic Variants of Uncertain Clinical Significance,” *Annals of Internal Medicine*, vol. 175, no. 7, Apr. 2022, doi: https://doi.org/10.7326/m21-4109.